

GameHorizon Suite: Multi-Horizon Data and Evaluation in Gameplay

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<https://arxiv.org/abs/2609.25001>



Annotator pipeline: from raw traces to standardized benchmark clips and questions.

Abstract

Modern video games provide a measurable testbed for AI models, combining abilities of visual understanding, instruction decomposition, goal planning, and precise action control over multiple temporal horizons. Existing datasets and benchmarks, however, either cover a narrow range of games, lack language instructions, or rely on high-variance online rollouts. To address these challenges, we introduce GameHorizon, a unified data and evaluation suite that measures game-play capabilities at different horizons for diverse model families. GameHorizon Suite consists of three components. First, GameHorizon-Annotator is a scalable and automated annotation pipeline for multi-horizon instructions. Second, utilizing the pipeline, we construct GameHorizon-Data, the first large-scale AAA gameplay dataset with temporally aligned videos, player actions, and multi-horizon instructions. It comprises 5,000 hours of recordings from 21 games, collected by 100 human expert players. Third, we build GameHorizon-Bench with reproducible offline and stepwise online testing. The offline track enables reproducible evaluation using thousands of standardized questions organized into three primary tasks and a series of diagnostic variants, while the online track tests whether offline scores reflect actual gameplay capabilities and local-izes failures to specific steps within long-horizon gameplay. Based on our GameHorizon Suite, we evaluate 47 models through more than one million model invocations, revealing a meaningful hierarchy of task difficulty and pronounced differences in model capabilities. Our work can provide a standardized yardstick for evaluating gameplay capabilities across horizons and model families. We will release our dataset, annotator, and benchmark to facilitate future research.

Task

Measuring Multi-Horizon Gameplay

- The paper introduces GameHorizon, a unified data and evaluation framework.
- The corpus comprises 5,000 hours of human gameplay across 21 game titles.
- The suite evaluates long-horizon gameplay through order-dependent causal tasks and order-flexible thematic tasks.
- Each task contains 2-6 verifiable subtasks to assess short-horizon capabilities.

Challenges

Narrow Coverage and Reproducibility Issues

- Existing datasets like GameWorld target only simple mini-games, resulting in narrow game coverage.
- Prior methods use small samples for low-confidence comparisons and online rollouts that are hard to reproduce.
- Earlier work focuses on Minecraft or non-AAA games.

Proposed method

Annotator Pipeline and Benchmark Design

- Action-aware segmentation uses keyboard-mouse traces to identify key action transitions and determine clip boundaries.
- Dynamic programming enforces duration ranges: 1–5 seconds for L1, 1–2 minutes for L2, and 5–8 minutes for L3.
- Unified multiple-choice questions ensure standardized evaluation where models directly output selections from four options.

Main results

Proprietary Models Lead, UMMs Lag

- Mean accuracy is 64.7%, substantially above the 25% random baseline.
- Proprietary model GPT-6-Astra ranks first with 80.2% overall accuracy.
- Best UMM BAGEL-7B-MoT achieves 61.4%, falling into Tier 3 or Tier 4.